1. C program to print a string.

```
#include <stdio.h>
#include <conio.h>
void main()
 {
  char greetings[] = "Hello World!";
  char blessings[]={'T','h','a','n','k',' ','Y','o','u','!','\0'};
  printf("\n%s\n", greetings);
    printf("%s", blessings);
getch();
}
```

2. C program to insert a character in a string

```
#include <stdio.h>
#include <conio.h>
void main()
 {
  char greetings[] = "Hello World!";
  greetings[5] = 'J';
  printf("%s", greetings);

getch();
}
```

3. C program to find the length of string using strlen() function.

```
#include <stdio.h>
#include <conio.h>
#include <string.h>
void main()
{
  char greetings[] = {'H', 'e', 'l', 'l', 'o', ' ', 'W', 'o', 'r', 'l', 'd', '!', '\0'};
  char greetings2[] = "Hello World!";

  printf("%d\n", sizeof(greetings));
  printf("%d\n", sizeof(greetings2));
  printf("%d\n", strlen(greetings));
  printf("%d\n", strlen(greetings2));

getch();
}
```

4. C program to copy one string into another using strcpy() function.

```
#include <stdio.h>
#include <conio.h>
#include <string.h>

void main()
{
  char str1[20] = "Hello World!";
  char str2[20];

  // Copy str1 to str2
  strcpy(str2, str1);

  // Print str2
  printf("\n%s", str2);
  // Print str1
  printf("\n%s", str1);
  getch();
}
```

5. C program to concatenate one string to another using strcat() function.

```
#include <stdio.h>
#include <conio.h>
#include <string.h>
void main()
{
  char str1[20] = "Hello ";
  char str2[10] = "World!";

  // Concatenate str2 to str1 (the result is stored in str1)
  strcat(str1, str2);

  // Print str1
  printf("\n%s", str1);
// Print str2
  printf("\n%s", str2);

getch();
}
```

6. C program to compare two strings using strcmp() function.

```
#include <stdio.h>
#include <conio.h>
#include <string.h>
void main()
{
char str1[] = "Hello";
  char str2[] = "Hello";
  char str3[] = "Hi";

  // Compare str1 and str2, and print the result
  printf("%d\n", strcmp(str1, str2));

  // Compare str1 and str3, and print the result
  printf("%d\n", strcmp(str1, str3));

  getch();
}
```

7. C program to reverse a string using strrev() function.

```
#include <stdio.h>
#include <conio.h>
#include <string.h>
void main()
 {
   char str[10];


   printf("\nEnter first string :");
   scanf("%s",str)  ;


   printf("\nOriginal String is %s ", str);
   printf("\nReversed String is %s ", strrev(str));

  getch();
}
```

8. C program to find the length of string without using library function.

```
#include <stdio.h>
#include <conio.h>

void main()
{
      char str[20];
   int i=0,length=0;

       printf("Enter a string: \n");
   scanf("%s",str);

   while(str[i]!='\0')
   {
       length++;     //Counting the length.
       i++;
   }

   printf("\nLength of input string: %d",length);

  getch();
}
```

9. C program to copy one string into another without using library function.

```
#include <stdio.h>
#include <conio.h>

void main()
 {
   char str1[10], str2[10];
   int i;

   printf("\nEnter first string :");
   scanf("%s",str1);

   i = 0;
   while (str1[i] != '\0')
     {
      str2[i] = str1[i];
      i++;
   }
   str2[i]='\0';

   printf("\nString1 is %s ", str1);
   printf("\nString2 is %s ", str2);

  getch();
}
```

10. C program to reverse a string without using library function.

```
#include <stdio.h>
#include <conio.h>

void main()
 {
   char str1[10], str2[10];
   int i,length=0;

   printf("\nEnter first string :");
   scanf("%s",str1);

       i = 0;
     while (str1[i] != '\0')
     {
      length++;
      i++;
   }
       i = 0;
   while (str1[i] != '\0')
     {
      str2[length-i-1] = str1[i];
      i++;
   }
   str2[i]='\0';

   printf("\nString1 is %s ", str1);
   printf("\nReversed String2 is %s ", str2);

  getch();
}
```

11. C program to check whether the given string is palindrome or not.

```
#include<conio.h>
#include<stdio.h>
#include <string.h>

void main()

   char string1[20];
   int i, length;
   int flag = 0;

   printf("Enter a string:");
   scanf("%s", string1);

   length = strlen(string1);

   for(i=0;i < length ;i++){
      if(string1[i] != string1[length-i-1]){
         flag = 1;
         break;
        }
      }

   if (flag) {
      printf("%s is not a palindrome", string1);
   }
   else {
      printf("%s is a palindrome", string1);
   }

 getch();
}
```